# Clarity before production.

SpecGuard AI compares manufacturing documents and connects each finding to its evidence.

## One order.
## Every detail.
## Human judgment.

Live inference via AMD Token Factory / MiMo-V2.6-Flash

# One order. Three conflicting sources.

An order coordinator needs to resolve differences before material is cut or printing begins.

### Buyer message

Custom hoodie

**300 pieces**
Color: **black**

### Technical PDF

Custom hoodie

**280 pieces**
Color: **charcoal**

### Size spreadsheet

S 75 / M 100
L 75 / XL 50

Total: **300 pieces**

**Three deliberate findings: quantity, color and size total. Synthetic data only.**

# From documents to a reviewed decision.

A working Streamlit interface for one product and one manufacturing order at a time.

### 01 / Upload

TXT, text-based PDF
and size CSV

### 02 / Inspect

Extracted facts with
quotes and locations

### 03 / Compare

Deterministic findings
and clear questions

### 04 / Review

Record valid corrections and reviewer notes.
Preserve original evidence.

### 05 / Export

Download JSON, CSV and clarification TXT.
Keep the order's review record.

# Live extraction. Independent checks.

AMD Token Factory inference performs a meaningful part of the working document workflow.

## Source parsing

TXT / pypdf / CSV
Retain line, page and record references

## AMD Token Factory

MiMo-V2.6-Flash
Authenticated live model requests

## Validation

Pydantic schema and evidence
Template value / coverage checks

## Reconciliation

Compare quantities, sizes, colors and manufacturing details

## Human decisions

Immutable original facts
Separate correction history

## Results

HOLD / REVIEW / READY
Evidence-linked reports

# Make uncertainty visible.

The app supports review; independent human production approval remains required.

## HOLD

Unresolved critical contradictions.

Example: 300 versus 280 pieces.

## REVIEW

Missing data, failed processing or unverified extraction coverage.

## READY

No blocking findings under the configured rules after valid corrections.

Free-form content cannot certify its own completeness. Unsupported layouts stay in REVIEW.
No OCR, multi-item orders, login or persistent signed audit storage in this prototype.

# Verified behavior, with a defined scope.

The repository separates deterministic regression results from live model smoke tests.

## 86 automated tests

Parsing, schema and source checks, false-READY regressions, correction safety and Streamlit flows.

## 27 synthetic orders

81 documents / 399 expected field values.
All expected statuses in deterministic Demo Mode.

## Hosted AMD smoke test

3 live documents verified; expected HOLD conflicts.
One observed order: 20.44 seconds end to end.

## Limits of this evidence

Synthetic templates, not unseen-document accuracy.
No customer savings or production SLA measured.

# Help order coordinators catch conflicts earlier.

Initial users: small and medium manufacturers receiving buyer messages, tech packs and size tables.

## Expected value

Less repetitive comparison.

Clearer clarification questions.

Traceable reviewer decisions.

## Pilot measurement

Compare review time and missed conflicts against manual work on independently labeled orders.

A future per-order fee or subscription is a hypothesis. Pricing, savings and customer demand remain unvalidated.

# Upload the examples. Inspect the evidence.

Suggested event track: Reinvent Commerce. Manufacturing quality also relates to Intelligent Industry.

## Hosted application

**https://specguard-ai-dev.streamlit.app/**

Choose Upload documents. Add buyer_order.txt, tech_pack.pdf and size_quantities.csv.
Extract and compare: expect HOLD with three deliberate conflicts.

## Code, examples and validation

https://github.com/minteeqahmed-png/Specguard-ai
MIT project code / examples folder / reproducible local Demo Mode